Public transit for all?

A Multi-Method Spatial Analysis of Transit Accessibility and Socio-Economic Patterns in Stockholm

Authors

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Data

Category	Variable	Data Required	Source	Purpose
Transit Density	Transit stops per 1,000 residents	GTFS stop locations + DeSO population data	Trafiklab GTFS Sweden + SCB	Dependent variable representing transit density
Socio-Economic Variable	Income per person	Income statistics per DeSO	SCB DeSO Statistics	To examine socio-economic influence on transit density
Behavioural / Mobility Variable	Cars per person	Registered vehicles per DeSO + population data	SCB Vehicle Statistics	Proxy for private mobility dependence
Urban Structure Variable	Distance from CBD	DeSO centroids + T-Centralen location	QGIS generated variable	To measure urban centrality

Methodology

Data collection
- Transit stops per 1,000 residents
- Income per person
- Cars per person
- Distance from CBD (T-Centralen)

Final variables
- Global Moran's I
- Univariate LISA
- Bivariate LISA

ESDA- *Exploratory spatial data analysis*

Spatial Error Model (SEM)

Spatial econometric analysis



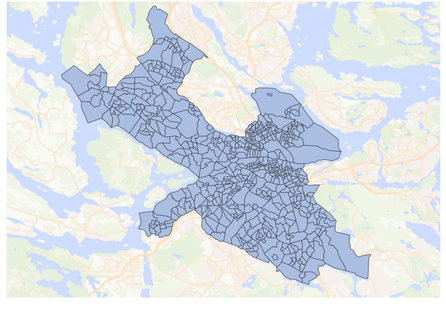
Data Processing
- Define site area
- Join DeSO boundaries
- Generate centroids
- Calculate Euclidean distance

Statistical methods
- Pearson Correlation
- OLS Regression

Interpretation
- Spatial clustering
- Socio-economic relationships
- Urban structure influence

Study area

The main spatial unit of analysis is the DeSO polygon, filtered to codes beginning with 0180 to isolate units within Stockholm municipality and link neighborhood-scale socioeconomic data with transport accessibility data.



Introduction

Stockholm is often viewed as a highly transit-oriented city with strong regional connectivity. However, transit density may still vary spatially across different urban areas. This study explores how transit density patterns relate to urban structure and socio-economic geography across Stockholm's DeSO units.The project also evaluates how spatial clustering and regression methods contribute differently to understanding urban accessibility patterns.

Hypothesis

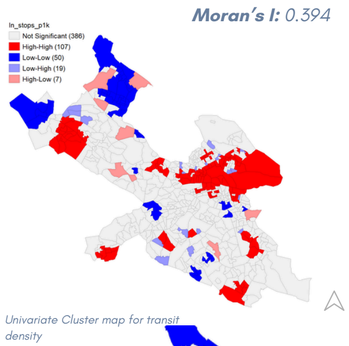
Spatial differences in transit density across Stockholm are associated with variations in socio-economic and urban characteristics.

Objective

- Identify spatial clustering patterns in transit density
- Examine relationships between transit density and urban variables
- Compare spatial and non-spatial statistical methods
- Understand the influence of spatial dependence in urban analysis

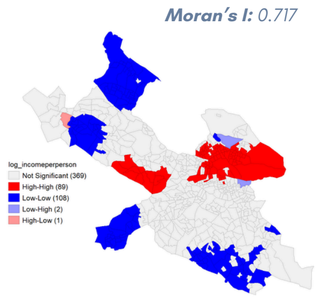
Results & Analysis

Cluster Analysis (Local Indicators of Spatial Association)



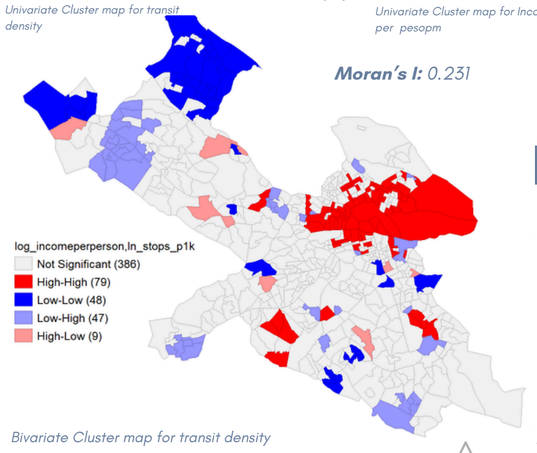
Moran's I: 0.394

Univariate Cluster map for transit density



Moran's I: 0.717

Univariate Cluster map for Income per pesopm



Moran's I: 0.231

Bivariate Cluster map for transit density

**All highlighted clusters are statistically significant at a minimum threshold of p < 0.05, with core urban and peripheral clusters reaching p < 0.01 and p < 0.001*

Key insights

Inner Core: Spatial Monopoly (High Income + High Transit)

Peripheral Areas: Double Disadvantage (Low Income + Low Transit)

Metro Corridors: Infrastructure Protection (Low Income + High Transit)
Tensta, Husby, Farsta, and Skärholmen

Affluent Suburbs: Choice Isolation (High Income + Low Transit)

Correlation

		ln_stops_p1k	dist_cbd	Carsperson	log_incomeperperson
ln_stops_p1k	Pearson Correlation	1	-.234**	.138**	.264**
	Sig. (2-tailed)		<.001	<.001	<.001
	N	569	569	569	569
dist_cbd	Pearson Correlation	-.234**	1	.367**	-.616**
	Sig. (2-tailed)	<.001		<.001	<.001
	N	569	569	569	569
Carsperson	Pearson Correlation	.138**	.367**	1	.311**
	Sig. (2-tailed)	<.001	<.001		<.001
	N	569	569	569	569
log_incomeperperson	Pearson Correlation	.264**	-.616**	.311**	1
	Sig. (2-tailed)	<.001	<.001	<.001	
	N	569	569	569	569

Key insights
- All variables significant at 0.01 level
- Income and distance from CBD strongest relationship **(r = -0.616)** - poorer areas are further out
- Income weakly predicts transit access **(r = 0.264)**

Spatial error model

Variable	Coefficient	p-value
Distance from CBD	-0.046	<0.001
Income per Person	-0.147	0.062
Cars per Person	2.068	<0.001
Model Performance	**Value**	
R^2	0.362	
Spatial Error λ	0.598	
Likelihood Ratio Test	p < 0.001	

OLS Regression

(Geoda classical regression model)

Variable / Statistic	Value	Significance
Distance from CBD	-0.041	p < 0.001
Income per Person	-0.064	p = 0.375
Cars per Person	1.672	p < 0.001
R^2	0.114	–
F-statistic	24.24	p < 0.001
Test	**Value**	**p-value**
Moran's I (error)	0.358	<0.001
LM Error	199.64	<0.001
Robust LM Error	15.31	<0.001

Key insights
- The OLS model showed **limited explanatory power (R^2 = 0.114).**
- **Income** was **statistically insignificant** in the classical model.
- Significant LM Error and Robust LM Error tests justified the use of SEM.

Key insights

Distance from CBD
As distance increases, Transit density decreases slightly

Income per Person
weak negative relationship with transit density and statistically not significant

Cars per Person
Higher car ownership aligned with higher transit density.

R^2
36.2% SEM substantially improved model fit over OLS.

Spatial error(λ)
Strong spatial dependence, indicating that neighbouring areas shared underlying geographic characteristics influencing transit density.

Conclusion

Spatial Patterns of Transit Density

- Transit density in Stockholm is not randomly distributed and forms clear spatial clusters around the urban core and major transit corridors.
- **Income is Insignificant (p = 0.062):** Lower-income peripheral districts remain highly connected via metro corridors, showing infrastructure distribution is shaped by intentional public planning

Role of Urban Structure

- Distance from the city center is the strongest predictor of transit density, reflecting a powerful center-periphery structure.
- High transit accessibility spans both affluent inner-city districts and dense, lower-income peripheral metro corridors.

Evaluation of methods

- ESDA and LISA were effective for identifying spatial clustering and revealing localized socio-spatial patterns.
- Correlation and OLS regression helped identify broader statistical relationships but failed to account for spatial dependence.
- The Spatial Error Model substantially improved model performance, demonstrating the importance of incorporating spatial effects in urban accessibility analysis.